Chapter 1. Generating Data

Sampling, experiments, randomization and blocking

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1.1 What is statistics?

Motivation. Opinion polls, drug trials and quality checks all try to say something about a large group after looking at only a small part of it. Statistics is the science of doing this in a way whose reliability can itself be assessed.

Table 1-1 shows the result of a poll taken before a mayoral election.

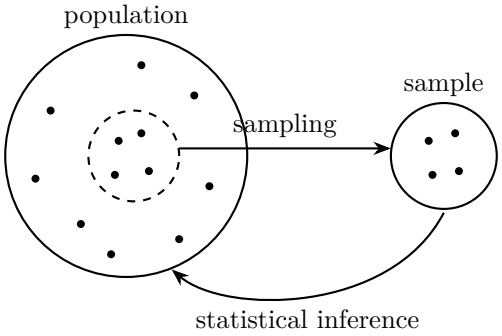
Candidate	Support
John Smith	42%
Emily Johnson	37%
Michael Lee	15%
Sarah Davis	6%

Such a poll raises three tasks. First, *whom* to survey: how many voters, chosen how, so that they represent all voters (including by age and occupation). Second, how to *organize and summarize* the data and draw conclusions about all voters—e.g. does Smith also lead Johnson in the whole electorate? Third, since the conclusion may be wrong, how to *assess its reliability*. **Statistics** is the science that collects, organizes and interprets numerical information (**data**) so as to reach rational conclusions with a measurable degree of reliability.

Usually only a tiny part of the units of interest can be observed: surveying everyone is too costly or impossible (one cannot record the sleep of every smartphone user in the world, nor give a new drug to every patient). So we collect data on a part and reason about the whole.

Definition (Population and sample)

Each individual unit making up the whole is a **sampling unit**; the value describing the feature of interest of a unit (support for a candidate, sleep quality, drug effect) is a **characteristic**. The collection of the characteristic values of *all* sampling units is the **population**; the collection of values actually observed is the **sample**.
A population made of finitely many units is a **finite population** (all voters of a city). A population with infinitely many units, e.g. all present and future smartphone users, is an **infinite population**; it is a conceptual, not a physically existing, object.



A brief history. The word "statistics" comes from *State Arithmetics*: rulers took censuses and valued land to levy taxes. In the 17th–18th centuries physics and astronomy demanded accurate measurement, and astronomers and surveyors met **variability**—repeated measurements of the same object differ—which stimulated statistical methods. In the 19th century statistics spread to agriculture and biology (heredity, comparison of crop varieties and fertilizers), relying on data from experiments and observation; in the 20th century it became a unified science. Today it is

used in epidemic forecasting, polling, medical data, quality control, autonomous-vehicle safety, AI evaluation, genetics, social-media analysis and climate research.

1.2 Simple random sampling

Motivation. Inference from a part to the whole is only trustworthy if the part represents the whole. The basic tool for guaranteeing this is to let chance, not people, pick the sample.

If a survey of the online-learning satisfaction of all Korean university students only covers students near the surveyor's home, the population is all students but the sample is not representative; if students in that region differ, the conclusion is wrong.

Illustration 1-1: a biased product sample

A factory makes thin steel plates for precision instruments. To test strength, ten $5\,\mathrm{cm}^2$ pieces are cut, but the inspector avoids visible defects and cuts only from good parts. The test then says the plates are excellent, although the sample does not reflect actual quality—customer complaints are likely.

Illustration 1-2: voluntary response

A singing contest let viewers vote by web, text or phone. Of 5 million viewers only 10% (500,000) voted; contestant A won with 50%. A later random survey of 500 viewers found only 10% preferred A and 90% preferred B: the voters were mostly A's fans. A sample made only of people who choose to respond is a **voluntary response sample**; it can badly distort the population's opinion.

Both illustrations show samples that misrepresent the population. A good sample gives every unit an equal chance of selection, which removes the surveyor's subjective intervention (Illustration 1-1) and the respondents' self-selection (Illustration 1-2), the sources of structural error. This requires careful *sample design*.

Definition 1-1 (Simple random sampling)

A sampling method designed so that, from a finite population, every subset consisting of n sampling units has the same probability of being selected as the sample is called **simple random sampling**.

If the population has size N, there are $\binom{N}{n}$ subsets of size n, so each has selection probability $1/\binom{N}{n}$. Concretely: pick the first unit with probability $1/N$; after k units have been chosen ($k = 1, \dots, n-1$), pick one of the remaining $N-k$ units with probability $1/(N-k)$. This is **sampling without replacement**: a chosen unit is not returned before the next draw. In **sampling with replacement** the chosen unit is returned, so each draw picks one of the N units with probability $1/N$ (*simple random sampling with replacement*); by contrast, *simple random sampling without replacement* picks with probabilities $\frac{1}{N}, \frac{1}{N-1}, \dots, \frac{1}{N-n+1}$.

Proposition (Properties of simple random sampling)

Simple random sampling (without replacement) of size n from a population of size N satisfies
(1) every possible sample of size n has the same chance of being selected;
(2) every unit of the population has the same chance of being selected, namely n/N.

Proof. [Supplement]

Step 1. With the sequential scheme, a particular ordered list of n distinct units is drawn with probability $\frac{1}{N} \cdot \frac{1}{N-1} \cdots \frac{1}{N-n+1}$. A given subset arises from $n!$ orderings, so its probability is

$$\frac{n!}{N(N-1)\cdots(N-n+1)} = \frac{n!\,(N-n)!}{N!} = \frac{1}{\binom{N}{n}},$$

the same for every subset; this is (1).

Step 2. A fixed unit belongs to $\binom{N-1}{n-1}$ of the $\binom{N}{n}$ equally likely subsets, so it is selected with probability $\binom{N-1}{n-1}/\binom{N}{n} = n/N$. ■

Systematic sampling (choose a random start, then every k-th unit) satisfies (2) but not (1). Statistical sample designs are based on **probability sampling**, in which the selection probability of each unit is explicitly defined; other designs are treated in Chapter 12.

1.3 Cautions in sample surveys

Motivation. Random selection removes the surveyor's bias, but a real survey can still go wrong in several practical ways. Recognizing them is as important as the sampling formula.

(1) Incomplete frame. One needs an accurate, complete list (frame) of the population's units. Homeless people, prisoners and dormitory residents may be missing from a household list; households without phones are missing from a telephone poll. If the omitted units differ from the rest, the sample suffers **bias**.

(2) Non-response. Selected people may refuse or be unreachable; non-response rates often reach 30%, and are higher in cities. Replacing non-respondents by extra respondents does not help if respondents and non-respondents hold different views.

(3) Response and interviewer effects. For sensitive questions ("Have you ever downloaded illegally?") people answer untruthfully, so the frequency is underestimated. An interviewer's tone, attitude or sex also matters; "Most people exercise three times a week; how often do you?" invites over-reporting. Interviewers must be trained and supervised.

(4) Question wording. The wording strongly steers answers. A question such as "Environmental regulation places an excessive burden on small firms, costing jobs and hindering growth. Do you favour relaxing it?" induces support for deregulation.

Reliable surveys therefore need a statistical sample design *and* careful questionnaires, a complete frame, trained interviewers and management of non-response.

1.4 Statistical experiments

Motivation. Surveys observe units as they are; experiments deliberately act on units and watch the response. This section fixes the vocabulary of experiments.

An experiment applies a manipulation to people, animals or objects and studies how they respond.

Definition 1-2 (Experimental unit and treatment)

The individual on which an experiment is performed is an **experimental unit**; applying a specific experimental environment or condition to an experimental unit is a **treatment**.

Definition 1-3 (Response variable, factor, factor level)

In a statistical experiment, a variable expressing the experimental environment or condition is a **factor**, and the variable expressing the reaction to it is the **response variable**. The values a factor takes are the **levels** of the factor.

Illustration 1-3: vaccine dose

The same dose of a vaccine is injected into 25 people and the blood concentration 30 minutes later is the response variable. The experiment has one factor (vaccine) at one level; with three different doses it would have one factor at three levels.

Example 1-1 (Diabetes treatment)

To study the effect of a new diabetes treatment on blood-sugar control, half of 18,000 participating patients take the new treatment daily and the other half the existing one. After six months the group on the new treatment has a lower mean blood sugar. Identify the experimental units, factor, levels and response variable.
Solution. Experimental units: the 18,000 diabetic patients. Factor: type of treatment (the experimental condition). Levels: two—new treatment and existing treatment. Response variable: blood sugar measured after six months.

Illustration 1-4: two factors

A textbook author asks whether practice problems work best at the start or at the end of a chapter, depending on the problem type. Six versions of the textbook are made and students get the same study time, homework and final test.

Position \ Type	Simple fact	Computation	Concept
End of chapter	Group 1	Group 2	Group 3
Start of chapter	Group 4	Group 5	Group 6

Students are the experimental units and the test score is the response. There are two factors, position (2 levels) and type (3 levels); their $2 \times 3 = 6$ combinations are the six treatments.

1.5 Comparative experiments and randomization

Motivation. A single-treatment experiment cannot separate the treatment's effect from everything else that happens to the units. Comparison with a control group and random assignment are the two remedies.

Illustration 1-5: gastric freezing

Gastric freezing—a patient swallows a balloon through which a coolant circulates for an hour—was claimed to reduce acid and relieve ulcer pain. The original study was

$$\text{observe pain} \;\rightarrow\; \text{gastric freezing} \;\rightarrow\; \text{observe pain}.$$

Patients improved, and the method was used for years. The improvement, however, came from trust in the doctor and expectation of the treatment: the **placebo effect**, a positive response to any treatment, which affects not only subjective symptoms but even physiological responses such as acid secretion. A later controlled study split 160 ulcer patients into 82 who

received gastric freezing and 78 who received a saline solution: 34% of the former and 38% of the latter improved.

The first study used one treatment only and was misleading. In the second, both groups received the placebo-type influence, so the difference between them is the effect of gastric freezing itself—which turned out to be nil. The group receiving the treatment is the **treatment group**; the group receiving the dummy treatment is the **control group**. Controlling extraneous factors through a control group is the core of experimental design.

First principle of experimental design

Minimize the effect of extraneous factors other than the factor of interest.

A control group alone is not enough. If the doctor knows who is truly treated, this can influence the assessment of pain; so neither patient nor experimenter should know the assignment: a **double-blind experiment**. Moreover, the two groups must be alike with respect to extraneous factors. If one group is more affected by an extraneous factor, the difference between groups mixes the treatment effect with that factor: this is **confounding**.

Illustration 1-6: confounded study

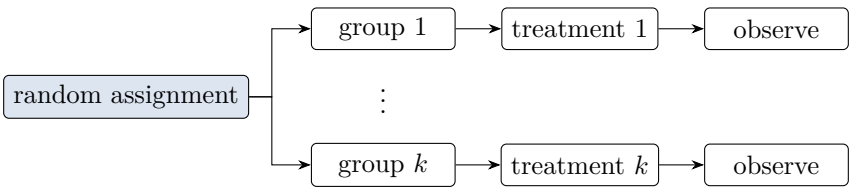
After toxic chemicals leaked near an industrial area, an agency reported genetic damage in residents, and the government decided on compensation. A review panel found the study had not controlled age or living habits, so damage could not be attributed to the chemicals. Genetic damage varies with age, sex and environment, and its assessment is subjective; residents should have been compared with a control group similar in age, sex and region, measured identically, and assessed blind.

In general there are too many extraneous factors to equalize them by hand, and an experimenter assigning patients by judgement (e.g. giving the new cancer drug to milder cases) can unintentionally bias the result. Hence units should be assigned purely by chance.

Illustration 1-7: random assignment

A food company evaluates a new instant food on 30 newly weaned male rats: 10 (treatment group) get the new food and 20 (control) the existing food, and weight gain after 4 weeks is measured. The 10 treated rats are chosen from the 30 by simple random sampling.

Each rat has the same probability 1/3 of being treated. Assigning experimental units by chance so that each has the same chance of each treatment is **randomization** (introduced by R. A. Fisher at Rothamsted).



Completely randomized design (Figures 1-2 and 1-3; $k = 2$ is the randomized comparative experiment)

Second principle of experimental design (randomization)

When deciding which treatment each experimental unit receives, give every unit the same chance of receiving any particular treatment.

When randomized experiments were proposed in the 1920s, many experimenters resisted, trusting their own fair judgement and fearing that chance might ruin a result. Experience showed the opposite: subjective assignment produces more unexpected errors, while random results can be predicted and explained by the laws of probability. "Random" is not "haphazard": in Illustration 1-7, taking the first ten rats caught is haphazard, and easily caught rats may be smaller or slower, biasing the comparison.

With three or more treatments, randomization works the same way: e.g. 30 units into three groups of 10 by drawing 10 by simple random sampling for group 1, 10 of the remaining 20 for group 2, and the rest for group 3. Dividing all units into as many groups as treatments and assigning by randomization is a **completely randomized design**.

Even with randomization, a difference in response is not automatically a treatment effect: in Illustration 1-7 rats grow at different rates, and fast growers might by chance cluster in one group. Randomization lets us judge by probability theory whether the difference is too large to be explained by chance (Chapter 6). Also, with one rat per group nothing can be concluded; using more units makes the groups more alike on average and the treatment effect easier to detect.

Third principle of experimental design (replication)

Apply each treatment repeatedly to a sufficiently large number of experimental units.

1.6 Blocking

Motivation. Complete randomization ignores known differences between units. When such differences are large, grouping similar units first and randomizing within groups gives far more precise comparisons.

Illustration 1-8: paired design

A psychologist studies the effect of room temperature (15 °C vs. 20 °C) on manual dexterity, measured by the number of pegs inserted into small holes in 30 minutes. A completely randomized design would split subjects into two groups, one per temperature; but individual dexterity varies so much that a small temperature effect may be hidden. Better: let every subject work at both temperatures, on different days (to avoid fatigue), with the order randomized (e.g. heads ⇒ 15 °C first) to neutralize learning and boredom.

This is a **randomized paired comparison design**: it keeps comparison, randomization and replication, but each unit receives both treatments and randomization decides the order. Each unit acts as its own control, removing between-person variation, so a small number of subjects suffices. Pairing two subjects of similar dexterity and randomizing which one works at 15 °C is also a paired design.

Definition (Blocking)

Blocking means forming, before the experiment, groups of experimental units expected to respond similarly to the same treatment; each such group is a **block**. A design that uses blocking and randomization (within blocks) together is a **randomized block design**; the randomized paired comparison design is a special case (block size 2).

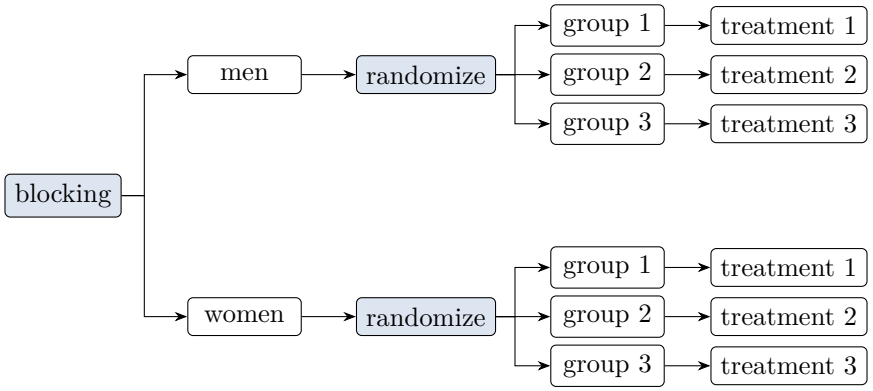
Blocks are built from factors expected to affect the response, so that their effect is controlled. Usually the block size equals the number of treatments (Illustration 1-9); sometimes blocks are larger and randomization splits each block into as many groups as treatments (Illustration 1-10).

Illustration 1-9: field blocks

To compare yields over 2 soybean varieties × 3 pesticides, fields differing in soil and fertility are each taken as a block and divided into 6 plots; within each block the 6 treatments are assigned to plots by randomization.

Illustration 1-10: blocking by sex

A cancer progresses at different speeds in men and women. A trial of three treatments uses 30 male and 30 female patients as two blocks; within each block, randomization forms three groups of 10, each receiving one treatment, and survival is compared within each block.



Randomized block design of Illustration 1-10 (Figure 1-4)

Whether by survey or experiment, the generated data are treated in analysis as a *sample from a population*. Often the population is a conceptual infinite population: in Illustration 1-9, for each variety–pesticide combination one imagines the infinite population of yields if that combination were applied to all fields, and the difference in mean yield is a difference between these six infinite populations.

Finally, a key contrast. Studying a drug's effect on blood pressure, an *experiment* lets the experimenter decide who takes the drug, so extraneous factors can be controlled; in a *sample survey* the units decide themselves and the surveyor merely observes, so control is hard. Experiments are not always possible: forcing people to take an unproven drug or to smoke is unethical. Then an **observational study** based on survey data is needed, and hidden extraneous factors must be interpreted with special care.

1.7 Selected exercises

Motivation. The exercises make the chapter's principles precise: why systematic sampling is not simple random sampling, how blocking and confounding decide whether an experiment can answer its question, and why randomization with replication balances extraneous factors.

Exercise 1.5 (Systematic sampling)

In systematic sampling the units are listed and selected at a fixed interval. For instance, to choose 4 addresses from a list of 100, the interval is $100/4 = 25$; a number r is drawn at random from $1, \ldots, 25$, and the sample is the addresses numbered r, $r + 25$, $r + 50$, $r + 75$ (for $r = 13$: $13, 38, 63, 88$).
(a) Describe how to select a systematic sample of size 5 from a list of 200 addresses.

(b) Show that every address has the same chance of being selected, but that the method is not simple random sampling.

Solution.

(a) The interval is $k = 200/5 = 40$. Draw r at random from $\{1, \dots, 40\}$, each value with probability $1/40$, and take the addresses $r,\ r+40,\ r+80,\ r+120,\ r+160$.

(b)

Step 1. The 40 possible samples $S_r = \{r, r+40, \dots, r+160\}$ are disjoint and together cover all 200 addresses, so each address j lies in exactly one of them (the one with $r \equiv j \pmod{40}$, $1 \le r \le 40$). Hence $P(j \text{ selected}) = 1/40 = 5/200 = n/N$, the same as under simple random sampling (property (2) of the proposition in Section 1.2).

Step 2. Simple random sampling gives each of the $\binom{200}{5} \approx 2.5 \times 10^9$ subsets of size 5 the same positive probability. Systematic sampling gives probability $1/40$ to each of the 40 sets S_r and probability 0 to every other subset; for example, addresses 1 and 2 can never be selected together. So property (1) of that proposition fails, and systematic sampling is not simple random sampling.

The practical consequence: if the list has a period that matches the interval (say every 40th address is a corner flat), all units of the sample are of the same type, and the estimate is biased. ■

Exercise 1.9 (Fungus and wind)

An experiment compares whether fungal disease spreads differently among three crops A, B and C. Fungal spread is known to be affected by the wind direction. The field is divided into three adjacent strips in a row along the wind direction, and A, B and C are planted in the three strips, C being nearest to the side from which the wind blows. Point out the problem with this experiment and propose a suitable design.

Solution.

Step 1. (The problem.) Each crop occupies one strip, so crop and position along the wind are completely *confounded*: C is always nearest to the wind and receives the most spores, A the fewest. Any difference in infection may be due to the crop or to the position, and the data cannot separate the two. Moreover each crop is grown on a single plot, so there is no replication and no estimate of plot-to-plot variability.

Step 2. (A design.) Exposure to the wind depends on the position along the wind direction. Divide the field into several *blocks*, each a band lying across the wind (perpendicular to it), so that all plots in a block are about equally exposed. Split each block into three plots and assign A, B and C to them at random, independently in each block.

Step 3. (Why it works.) This is a randomized block design (Section 1.6). Every crop appears once in every block, so all crops meet the same range of wind exposure; comparisons are made within blocks, which removes the exposure differences; the within-block randomization protects against remaining unknown differences between plots; and the several blocks provide replication. ■

Exercise 1.11 (A confounded comparison)

To test whether a new learning programme is more effective than the current one, a school randomly splits the 40 students of one class into two groups of 20. One group studies with the new programme, taught by one teacher; the other group studies with the current programme, taught by another teacher. At the end the new-programme group has improved more, and the school concludes that the new programme is effective. Although the students were randomized, the experiment has a serious defect. Explain why the conclusion cannot be

trusted.

Solution.

Step 1. The treatment actually applied to a group is "programme together with teacher". Each programme is taught by exactly one teacher, so the programme effect is confounded with the teacher effect: a better or more enthusiastic teacher would produce the same result with no programme effect at all. Randomizing the students balances the students, not the teachers.

Step 2. There is no replication at the level at which the programmes differ: one teacher per programme gives no information on how much results vary between teachers, so even a large difference cannot be judged against teacher-to-teacher variation.

Step 3. A sound design separates the two factors: for example, let each teacher teach both programmes to randomly assigned halves of their students (teachers as blocks), or randomize several classes and several teachers to the programmes. Then the programme effect is estimated by comparisons in which the teacher is held fixed or balanced by randomization. ■

Exercise 1.13 (Randomization balances a covariate)

There are $2N$ experimental units with values $u_1, \dots, u_{2N}$ of some extraneous variable (age, fertility, ...), with mean $\bar{u}$ and $\sigma^2 = \frac{1}{2N}\sum_k (u_k - \bar{u})^2$. A completely randomized design sends a simple random sample of N units to treatment and the rest to control. Let $\bar{u}_T$ and $\bar{u}_C$ be the mean values of u in the two groups. Show that

$$E(\bar{u}_T - \bar{u}_C) = 0, \qquad \operatorname{Var}(\bar{u}_T - \bar{u}_C) = \frac{4\sigma^2}{2N-1}.$$

Solution [Supplement].

Step 1. Let $\delta_k = 1$ if unit k is treated and 0 otherwise, and $d_k = u_k - \bar{u}$, so that $\sum_k d_k = 0$ and $\sum_k d_k^2 = 2N\sigma^2$. Since $\sum_k \delta_k = N$ and $\bar{u}_C = 2\bar{u} - \bar{u}_T$,

$$\bar{u}_T - \bar{u} = \frac{1}{N}\sum_k \delta_k u_k - \bar{u} = \frac{1}{N}\sum_k \delta_k d_k, \qquad \bar{u}_T - \bar{u}_C = 2(\bar{u}_T - \bar{u}).$$

Step 2. By property (2) of the proposition in Section 1.2, $P(\delta_k = 1) = N/(2N) = \frac{1}{2}$. A fixed pair of units belongs to $\binom{2N-2}{N-2}$ of the $\binom{2N}{N}$ equally likely treatment groups, so for $k \neq l$

$$P(\delta_k = \delta_l = 1) = \frac{N(N-1)}{2N(2N-1)}, \qquad \operatorname{Cov}(\delta_k, \delta_l) = \frac{N-1}{2(2N-1)} - \frac{1}{4} = -\frac{1}{4(2N-1)},$$

while $\operatorname{Var}\delta_k = \frac{1}{4}$.

Step 3. (Mean.) $E\sum_k \delta_k d_k = \frac{1}{2}\sum_k d_k = 0$, so $E(\bar{u}_T - \bar{u}_C) = 0$.

Step 4. (Variance.) Using $\sum_{k\neq l} d_k d_l = (\sum_k d_k)^2 - \sum_k d_k^2 = -\sum_k d_k^2$,

$$\operatorname{Var}\Big(\sum_k \delta_k d_k\Big) = \frac{1}{4}\sum_k d_k^2 + \frac{1}{4(2N-1)}\sum_k d_k^2 = \frac{2N}{4(2N-1)} \cdot 2N\sigma^2 = \frac{N^2\sigma^2}{2N-1}.$$

Hence $\operatorname{Var}(\bar{u}_T - \bar{u}) = \sigma^2/(2N-1)$ and $\operatorname{Var}(\bar{u}_T - \bar{u}_C) = 4\sigma^2/(2N-1)$.

So randomization makes the two groups alike in u *on average*, whatever u is (even if it was never measured), and the size of a chance imbalance shrinks like $1/N$: this is how the second and third principles (randomization and replication) work together. Blocking on u removes the imbalance in u altogether. ■